

Halfway: Towards a Warmer Neighborhood

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ABSTRACT

Nowadays, people living in cities face a growing social isolation context. Following the global COVID-19 pandemic, contactless habits have grown as a sort of new normal for some people, diminishing the number of opportunities for communication. People are less likely to go towards other people to ask for help, including their neighbors, no matter how physically close they are. Halfway, as an apartment building-interconnected screen system, aims to nudge people to talk to their neighbors through services and games.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods; • User-centered design;

KEYWORDS

Social isolation, Neighbor relationships, Urban life

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1 INTRODUCTION

The United Nations aim to maintain international peace and security, promote economic and social development, protect human rights, and foster cooperation among nations to address global issues. The “Halfway: Towards a Warmer Neighborhood” project specifically aligns with the 11th UN Sustainable Development Goal, titled “Sustainable Cities and Communities”. This goal aims for cities to become safer and more inclusive and encourage green economies and resilient societies. The Halfway project takes on the social aspects of the goal, on an interpersonal level, by tackling social relationships in apartment buildings, and by using technology to bring people together in the physical world.

By its nature, Halfway also contributes to elements of Goal 3, “Good Health and Well-Being”, as it provides an help out of social isolation, thus promoting better mental health. It also contributes to Goal 10, “Reduced Inequalities”, as social isolation is a risk factor for income and social inequity.

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2 SOCIOPOLITICAL CONTEXT

Social isolation is a context in which a person finds themselves alone against their will, physically and/or emotionally [19]. A long-term and/or repeated isolation may impact a person’s health and security [9, 18]. Given the harmful and growing consequences of global isolation, the World Health Organization declared isolation as an urgent threat to health in 2023 [20].

For a long time now, people have been moving to big cities for various reasons, often finding themselves alone and far away from their close ones, which is conducive to isolation [1, 17]. The COVID-19 pandemic has revealed cracks in social systems and exacerbated this long-standing issue. The prolonged periods of social isolation it imposed have particularly weakened the mental health of the general population [21]. Physical human interactions significantly decreased, and several digital advancements that occurred due to safety measures prohibiting human contact remain part of people’s daily lives.

The consequences of this isolation are particularly felt in large cities. For example, Toronto, the most populous city in Canada, has been identified as the city with the highest amount of social isolation [16].

Combined with other contexts, for example of an economic nature, isolation can lead to consequences beyond mental health. It is notably a significant pathway to homelessness and may prevent someone in need from getting assistance [15].

3 PROCESS AND METHODS

To initiate the design process, methods inspired by ethnographic field research were used. Isolation, even when limited to urban populations, can be a complex issue, and the methods helped narrow down the numerous ways to approach it. Inspiration was drawn from the IDEO methods cards [5] and from the book “Universal Methods of Design” [11]. Additionally, documentary research on social isolation and ways it has previously been addressed has been conducted.

3.1 Graffiti Walls

To obtain initial general insights into the isolation of people living in cities, the Graffiti Wall method [11] was chosen. This method consists of placing a sheet on a public wall, accompanied by a pencil, to encourage people to write down their thoughts anonymously. In this case, the sheets were placed on the walls with pre-written prompts related to social isolation.

A total of five Graffiti Walls were installed in two buildings at Université Laval, located in two different districts of Quebec City. One building is one where classes are dispensed, and the other one houses students. The administration of both buildings gave express authorization for the Graffiti Walls to be installed. They all were placed in bathrooms for better privacy and were left in place for one month. Pictures were taken regularly during that time to



Figure 1: Overview of customer journey maps created during autoethnography research.

avoid losing data from tampering or vandalism and document the evolution.

This method led to two main insights. First, providing an anonymous tool for expression allows people to liberate themselves from deeper thoughts, seeing as the number of writings grew rapidly every day. Some of them were of a humorous nature, but most of the content adhered to the prompt and assumed a confessional tone. This led to believe that isolation may be a difficult subject to openly discuss for some people. Second, participants seemed inclined to mutual aid and compassion. This insight stemmed from the fact that the Graffiti Walls became platforms for dialogue from the day of installation. People replied with messages of solidarity, such as “You’re not alone in this”, “People do care”, or “That’s how I feel too”. On several occasions, participants gave advice or even left help resources’ phone numbers in response to darker messages. We noted that the cover of anonymity might facilitate not only requests for help but also interpersonal support.

3.2 Autoethnography-inspired Methods

In parallel to the evolving Graffiti Walls, we, the authors, took part in social contexts to gain perceptions and personal experiences regarding socialization, inspired by autoethnography methods [12]. First, we individually experienced an activity we would have been more comfortable doing accompanied. We chose to have a Saturday restaurant meal, when the experience is usually seen as more social than culinary [3].

We journaled our thoughts and feelings regarding the experiment from the beginning of the day, through the dining experience and after. We each mapped our experience through a customer journey map (Figure 1) with common pre-established factors and parameters [7].

Initially, we aimed to observe the behaviors of employees and customers around us in the face of our solitude. However, comparing our experiences, notes, and customer journey maps, we realized that for all of us, the experience with employees did not differ significantly from previous restaurant experiences, and the customers eating nearby were simply focused on their own social activity. The

main common tangent was the apprehension that accompanied the meal, starting several hours before even leaving the house. All three of us experienced enough apprehension to consider not going, but decided to go despite the stress, for the sake of our research. For all three, the experience went smoothly and was considered worth repeating later if the opportunity arose. We concluded that apprehension related to social norms may be a determining road-block to socialization, and that extrinsic motivation could strip these barriers.

Next, we participated in another social restaurant situation, as a group this time, to observe certain socialization behaviors. We assembled a group of about twenty people, that we mainly knew in a professional context, to observe and understand places conducive or not to isolation by integrating directly into people’s daily lives and observing their reactions to some contexts [12]. Initially, there were few exchanges; the conversation quickly fizzled out, leading to moments of silence. Then, a simple board game was placed on the table, which imposed a topic of discussion. People around the table asked questions about the rules, and discussions became fluid, even after the game was over. From this experience, we concluded that a structured context favors socialization.

3.3 Interviews and Group Discussions

Informal interviews were conducted with a total of seven people who lived in apartments. The interviewees were between the ages of 20 and 35 and comprised students and workers. Some lived on their own, others in shared flats, and in one case, with their child. The individuals we spoke to were not severely isolated; however, they expressed similar statements to those written on the Graffiti Walls. For example, they could have a fairly complete social life but feel isolated at various levels at home. Also, none of the interviewees knew their building neighbors, despite their proximity. Relationships, when they existed, were mostly characterized by a certain animosity, mainly motivated by noise. For instance, the people we spoke to mentioned feeling like their upstairs neighbor “deliberately stomped their feet as loudly as they could” or receiving complaints they deemed “excessive” from their downstairs neighbors. From these interviews, we noted the dichotomy between the constant pursuit of the isolation cycle, despite the physical proximity between people.

4 DESIGN OPPORTUNITIES

The main insight that came out of the research is that a lot of apprehension accompanies the idea of initial contact with a new person; people are afraid of bothering others and being judged. However, once the ice is broken, the apprehension decreases, and communication becomes easier with each interaction. To break the ice, people may feel like they need “permission” which can be granted by the context. For example, a work or school context provides this permission, along with a clear topic to start the discussion. Based on notes on a past conference by INSPQ [6], mutual aid was explored as a context that could provide this permission on a day-to-day basis.

5 RELATED WORK

At different stages, the Halfway project drew inspiration from mechanics already found in the market: for example, dating apps that operate on a “swiping” basis, where two users must mutually give permission before they can send messages to each other. In line with the findings on apprehension, it was judged relevant to integrate a similar concept of “reciprocal authorization” to reassure users in an apartment life context. Additionally, documentation on previous digital efforts tackling social isolation and anxiety [13, 14] helped frame our research and design ideas.

During our design activities, we found out about another project that tackled similar goals. It is a mobile application called Nextdoor, which serves as a social network to bring together local communities by creating groups with common interests and facilitating the sale of items between individuals. Finding out about Nextdoor allowed us to confirm some design choices we had taken following our ethnographic and documentary research activities. It turned out that the Nextdoor application had been used in some neighborhoods in the United States as a tool for racial profiling [2]. Early in the project, we chose to not include personalized communications in favor of integrated features. One of our motivations was the number of testimonies from participants who had rather tense relationships with their neighbors. We wanted to avoid Halfway being used in a hostile manner.

6 DESIGN SOLUTION

The proposed solution is named “Halfway,” and provides the first step in socializing among neighbors within an apartment building. It allows the first step to be taken anonymously, thus reducing the apprehension associated with face-to-face social interaction [8].

As seen on Figure 2, as a physical object, Halfway is a wall-mounted touchscreen. For better usability, the screen is detachable, and the stand serves the dual function of storing and charging. In a building, each unit is equipped with its own Halfway screen; the screens are interconnected, allowing system exchanges between residents. The installation is provided by the property owner with the assistance of electricians if needed. Once removed from the stand, the screen operates on battery power. Additionally, the stand is customizable; the property owner can choose its materials and colors.

There are two main functionalities on Halfway: mutual aid, and a game room.

6.1 Mutual Aid

The mutual aid functionality allows residents to ask for help related to daily tasks. Possible features include asking for an ingredient, borrowing a tool, or donating goods. For example, if a resident is missing an ingredient for a recipe, they could ask the other people living in the building to provide it to them through Halfway.

The mutual aid features work on a “reciprocal authorization” basis. The resident asking for a favor would first use the interface to select a category (ingredient, tool or donation). Inside a category, they could choose a precise request from the menu, or search it through the search bar. For example, in the case of an ingredient, the resident could select “milk”, then submit the request for a neighbor to provide some to them. Once the request is submitted, all the



Figure 2: The screen can be removed from its stand by sliding it towards the side slot. The shape of the stand resembles a comic book speech bubble, referencing the communication goal of the system. The tail of the bubble also facilitates the sliding motion by providing an open access to the bottom of the screen.

other residents in the building receive a notification on their own Halfway screen.

As seen on Figure 3, the request reads that “a neighbor” placed the request. The choice to anonymize the notification aims to reduce the apprehension related to placing a request to a minimum: in the event where nobody responds, the resident would be less likely to relate the lack of an answer to their personal value and image.

When a resident receives a notification, the Halfway stand lights up with a neon outline, additionally to the screen notification. To follow the “reciprocal authorization” principle, the responding resident’s answer to the request is also anonymous. Following the response, the interface provides instructions to both residents to follow up on the exchange in a common area of the building. Once a response is sent, the notification is instantly removed from the other residents’ screens.

Exchanges are entirely button-based, without allowing typed-in text apart from the search bar. This was a deliberate choice, following feedback received during the informal interviews regarding tense relationships between neighbors. Removing altogether the personalized message feature is a way to prevent Halfway from being used to send hostile messages, which could be encouraged by the system’s anonymity [4].

These exchanges, by starting digitally and ending face-to-face, can ease neighbors into meeting and exchanging words for the first time, breaking the ice. Also, due to the collaborative context of the exchange, the neighbors may share a feeling of complicity towards each other, prompting further dialogue the next times they cross paths.

As a security measure, exchanges will always take place in a common area of the building. Apartment numbers are never disclosed in the process. Additionally, a history of a resident’s interactions using Halfway is kept on their device.

6.2 Game Room

On the playful side, Halfway includes a game room that grants access to mobile games to residents, playable as if on a tablet. Invitations to play would be sent out similarly to the mutual aid



Figure 3: Overview of the interfaces on the Halfway screen when receiving a Mutual Aid request from a neighbor. The first screen (from the left) is the initial notification, and transitions into the second screen after a few seconds. The third screen is what appears after the resident pressed the “Yes” button.

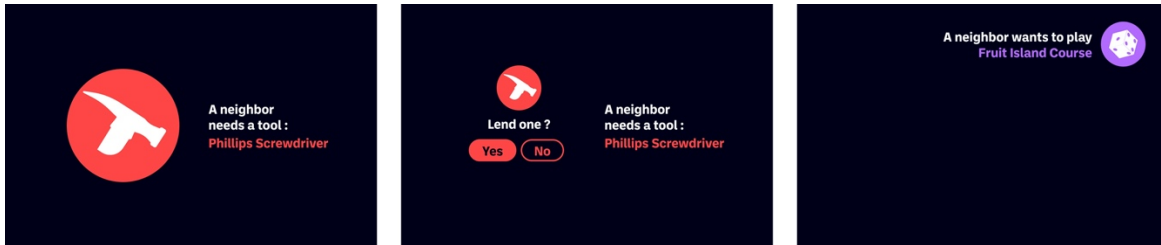


Figure 4: Interface comparison between the initial notification for a Mutual Aid request (two screens on the left) and a Game Room invitation (right).

functionality. The notification would be graphically organized differently on the screen, for a quick glance from across the room to be sufficient to differentiate a mutual aid request from a game invitation (Figure 4).

The game room adds to Halfway another way to gain awareness of the other residents in the building. This would give another way for neighbors to connect. Also, playing games together provides an additional contact and longer connections.

Specific games to include in the Halfway game room have not been chosen yet. The main criteria for inclusion would be for the games to be simple and turn-based so each resident can play at their own pace.

7 EVALUATION

To assess the feasibility of the Halfway project, evaluations were conducted in the form of discussion tables. The project was presented in detail to the participants, who were invited to react and discuss the potential of the concept and provide suggestions for improvement.

The project was generally well-received and understood by the discussion groups. Due to the high number of elderly individuals living in isolation, the accessibility challenge for this demographic was mentioned several times. Some participants expressed concerns that the digital nature of the project might alienate these users, leading them to feel rejected, even though they would benefit the most from a solution to break isolation.

While this concern is valid, an increasing number of elderly individuals have Internet connections at home and use electronic devices [10]. However, it will be important to keep the interface very simple in case some users face more adaptation challenges with

Halfway. Additionally, increased attention to accessibility should be evident in the design decisions of the interface for users with visual impairments.

Another noteworthy observation from the discussion groups compared Halfway to changes in neighborhood habits in the aftermath of the COVID-19 pandemic. Participants highlighted the decrease in neighborhood gatherings, even since sanitary measures were lifted, in line with our initial documentary research, and concluded that this solution would be relevant to rebuild these past habits. They also emphasized that the presence of such a device in residential buildings during lockdown periods would have been beneficial for maintaining certain connections, especially with the game room.

While the opinion of some participants aligned with our conclusions regarding apprehension and fear of judgment, stating that anonymity, using Halfway, would act as an incentive for them, others raised concerns about its possible lack of humanity. Some participants considered themselves less likely to be engaged by an anonymous message than by a name accompanied by a photograph. Although most participants were in favor of anonymity, we keep in mind the possibility of allowing users to customize a profile, within certain limits. However, such a feature would require further research and testing on its own.

Ultimately, participants concluded that the solution was relevant for today’s predominantly virtual context for easier real-life connections and would use Halfway, were it installed in their apartment building.

8 CONCLUSION

Given current trends, where the sense of isolation becomes stronger and relationships are increasingly challenging to form due to various issues, a lot of design opportunities arise to improve relationships among neighbors in a bustling city.

Drawing on ethnographic-inspired research and participatory observations, Halfway, a solution that allows residents of the same building to forge initial connections and recreate a sense of community among neighbors to reduce isolation, was created.

Halfway is an interconnected, user-friendly screen nudging apartment neighbors into meeting and helping each other. Unlike existing alternatives in the form of mobile applications, Halfway avoids the negative aspects of social networks by removing altogether text messages.

Evaluations and group discussions have demonstrated the solution's relevance for all generations while remaining accessible and integrated into residents' daily lives.

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At present, the Halfway project remains a fictive concept. More specific features, particularly in terms of customization, are yet to be tested and designed. A more comprehensive creation of interfaces will help clarify certain details relevant to user experience, ensuring a better growth for the project.

To test a real version of the project, we would consider partnering with a housing company that could integrate the project directly into one of its buildings and gather feedback from tenants. The accommodations equipped with the "Halfway" option would bring added value for both the landlords and the tenants who would benefit from it. The housing company would emphasize the improvement of the quality of life using these accommodations.

REFERENCES

- [1] Cédric Andrès. 2012. Culture de centre-ville et santé mentale: une réflexion anthropologique. *Santé mentale au Québec*. 37, 1 (October 2012), 93-101. <https://www.erudit.org/en/journals/smq/2012-v37-n1-smq0304/1012646ar/>
- [2] Andrew Anthony. 2022. Is media platform Nextdoor a friend in need or a vigilante nightmare? *The Guardian*. (March 2022). Retrieved November 22, 2023 from <https://www.theguardian.com/media/2022/mar/20/is-nextdoor-a-friend-in-need-or-vigilante-nightmare>
- [3] Yevvon Yi-Chi Chang. 2021. Lonely or alone? Solitary dining in Japan and Taiwan. *International Journal of Culture, Tourism and Hospitality Research*. 15, 1 (September 2021), 10–25. <https://doi.org/10.1108/IJCTHR-01-2020-0008>
- [4] Tony Cooper, Constantino Stavros, & Angela R. Dobe. 2019. Domains of influence: exploring negative sentiment in social media. *Journal of Product & Brand Management*. 28, 5 (February 19), 684–699. <https://doi.org/10.1108/JPBM-03-2018-1820>
- [5] IDEO. 2023. *IDEO Method Cards: 51 Ways to Inspire Design*. IDEO, San Francisco, CA.
- [6] Institut national de santé publique du Québec. 2018. L'isolement social, une épidémie évitable par l'entraide et les liens. (December 2018). Retrieved from <https://www.inspq.qc.ca/jasp/l-isolement-social-une-epidemie-evitable-par-l-entraide-et-les-liens-sociaux>
- [7] Jim Kalbach. 2020. *Mapping experiences* (2nd. ed.). O'Reilly Media Inc, Sebastopol, CA.
- [8] Ruogu Kang, Stephanie Brown & Sara Kiesler. 2013. Why do people seek anonymity on the internet? Informing policy and design. *Proceedings of the SIGCHI conference on human factors in computing systems*. ACM, Paris, 2013
- [9] Louise Hawkey & John T. Cacioppo. 2010. Loneliness Matters: A Theoretical and Empirical Review of Consequences and Mechanisms. *Annals of Behavioral Medicine*. 40, 2 (July 2010), 218–227. <https://academic.oup.com/abm/article/40/2/218/4569527>
- [10] Sonia V. M. Lozoya, Martha A. Z. Guirado, Alfredo Z. Gonzalez & Ana B. M. Lopez. 2022. Use of technologies and self-efficacy in older adults. *Ieee Revista Iberoamericana De Tecnologias Del Aprendizaje*. 17, 2 (May 2022), 125-130. <https://ieeexplore.ieee.org/document/9756012>
- [11] Bella Martin & Bruce Hanington. 2018. The pocket universal methods of design: 100 ways to research complex problems, develop innovative ideas and design effective solutions. Rockport, Beverly, MA.
- [12] Nicolas Nova. 2014. Beyond Design Ethnography: How Designers Practice Ethnographic Research. SHS, Genève, SWZ.
- [13] Maria Pugliese & Cordula Vesper. 2022. Digital joint action: Avatar-mediated social interaction in digital spaces. *Acta Psychologica*.
- [14] Radio-Canada. 2023. Quartier Québec, un repère en ligne divertissant pour briser l'isolement des jeunes. (April 2023). Retrieved from <https://ici.radio-canada.ca/nouvelle/1969509/quartier-quebec-repere-divertissant-briser-isolement-jeunes>
- [15] Jessica Rea. 2023. Social relationships, stigma, and wellbeing through experiences of homelessness in the United Kingdom. *Journal of Social Issues*. 79, 1 (November 2023), 465-493.
- [16] Toronto Foundation. 2022. Social isolation on the rise while civic engagement has dropped in Toronto, study finds. Retrieved November 20, 2023 from <https://torontofoundation.ca/socialisolationontherise>
- [17] W. Russell Tylor. 1931. The Exodus From Rural America. *Current History*. 35, 3 (October 1931), 404-408.
- [18] Laetitia Van Eeckhout. 2022. «L'urbanisme relationnel» au secours de la solitude des grandes villes. *Le Monde*. (November 2022) https://www.lemonde.fr/smart-cities/article/2022/11/26/l-urbanisme-relationnel-au-seccours-de-la-solitude-des-grandes-villes_6151715_4811534.html
- [19] Netta Weinstein, Heather Hansen & Thuy-vy Nguyen. 2022. Definitions of solitude in everyday life. *Personality and Social Psychology Bulletin*. 49, 12 (December 2023), 1635-1736. <https://doi.org/10.1177/01461672221115941>
- [20] World Health Organization. 2023. WHO Commission on Social Connection. Retrieved from <https://www.who.int/groups/commission-on-social-connection>
- [21] Naohiro Yonemoto & Yoshitaka Kawashima. 2023. Help-seeking behaviors for mental health problems during the COVID-19 pandemic: A systematic review. *Journal of Affective Disorders*. 323 (February 2023), 85-100.